

```
In [1]: #set operations
s1={1,2,3,4,5}
s2={4,5,6,7,8}
s3={8,9,10}
s1.union(s2)
```

Out[1]: {1, 2, 3, 4, 5, 6, 7, 8}

```
In [2]: s1.union(s2,s3)
```

Out[2]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

```
In [3]: s2|s3
```

Out[3]: {4, 5, 6, 7, 8, 9, 10}

```
In [4]: s1|s2|s3
```

Out[4]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

```
In [5]: print(s1)
```

{1, 2, 3, 4, 5}

```
In [6]: s1.intersection(s2)
```

Out[6]: {4, 5}

```
In [7]: s2&s3
```

Out[7]: {8}

```
In [10]: s1.difference(s2)
```

Out[10]: {1, 2, 3}

```
In [11]: s2-s3
```

Out[11]: {4, 5, 6, 7}

```
In [12]: print(s1)
print(s2)
print(s3)
```

{1, 2, 3, 4, 5}

{4, 5, 6, 7, 8}

{8, 9, 10}

```
In [14]: s1.symmetric_difference(s2)
```

Out[14]: {1, 2, 3, 6, 7, 8}

```
In [15]: s2^s3
```

Out[15]: {4, 5, 6, 7, 9, 10}

```
In [19]: s1.symmetric_difference_update(s2)
s1
```

```
Out[19]: {1, 2, 3, 6, 7, 8}
```

```
In [21]: #types of sets
a={1,2,3,4,5,6,7,8,9}
b={3,4,5,6,7,8}
c={10,20,30,40}
```

```
In [22]: a.issuperset(b)
```

```
Out[22]: True
```

```
In [23]: b.issubset(a)
```

```
Out[23]: True
```

```
In [25]: c.isdisjoint(a)
```

```
Out[25]: True
```

```
In [27]: a1 = {1,2,3,4,5,6,7,8,9}
b1 = {30,40,50,60,70,80}
c1 = {10,20,30,40}
```

```
In [28]: a1.issuperset(b1)
```

```
Out[28]: False
```

```
In [29]: b1.issubset(a1)
```

```
Out[29]: False
```

```
In [30]: a1.isdisjoint(b1)
```

```
Out[30]: True
```

```
In [32]: #dictionary
d={}
d
type(d)
```

```
Out[32]: dict
```

```
In [34]: d1={1:'one', 2:'two', 3: 'three'}
d1
```

```
Out[34]: {1: 'one', 2: 'two', 3: 'three'}
```

```
In [35]: d2 = {'one' : 1, 'two':2, 'three' : 3}
d2
```

```
Out[35]: {'one': 1, 'two': 2, 'three': 3}
```

```
In [36]: d2.keys()
```

```
Out[36]: dict_keys(['one', 'two', 'three'])
```

```
In [37]: d2.values()
```

```
Out[37]: dict_values([1, 2, 3])
```

```
In [38]: d2.items()
```

```
Out[38]: dict_items([('one', 1), ('two', 2), ('three', 3)])
```

```
In [39]: len(d2)
```

```
Out[39]: 3
```

```
In [40]: myset={1,2,3,4,5}#Set of numbersmyset  
myset
```

```
Out[40]: {1, 2, 3, 4, 5}
```

```
In [42]: len(myset)#Length of the set
```

```
Out[42]: 5
```

```
In [43]: myset={1,1,2,2,3,4,5,5}  
myset
```

```
Out[43]: {1, 2, 3, 4, 5}
```

```
In [45]: myset1={1.79,2.08,3.99,4.56,5.45}#Set of float numbersmyset  
myset
```

```
Out[45]: {1, 2, 3, 4, 5}
```

```
In [46]: myset2={'Asif','John','Tyrion'}# Set of Stringsmyset  
myset2
```

```
Out[46]: {'Asif', 'John', 'Tyrion'}
```

```
In [47]: myset3={10,20,'holla',(11,22,33)}  
myset3
```

```
Out[47]: {(11, 22, 33), 10, 20, 'holla'}
```

```
In [48]: myset3={10,20,"HoLa",[11,22,32]}# set doesn't allow mutable items like listmyset3  
myset3
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[48], line 1  
----> 1 myset3={10,20,"HoLa",[11,22,32]}# set doesn't allow mutable items like list  
t3  
      2 myset3  
  
TypeError: unhashable type: 'list'
```

```
In [51]: myset4=set()  
print(type(myset4))
```

```
<class 'set'>
```

```
In [53]: #Looping through set  
myset={'one','two','three','four','five','six','seven','eight'}
```

```
In [54]: for i in myset:  
        print(i)
```

```
six  
three  
seven  
four  
eight  
five  
two  
one
```

```
In [56]: for i in enumerate(myset):  
        print(i)
```

```
(0, 'six')  
(1, 'three')  
(2, 'seven')  
(3, 'four')  
(4, 'eight')  
(5, 'five')  
(6, 'two')  
(7, 'one')
```

```
In [57]: #set membership  
myset
```

```
Out[57]: {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [58]: 'one' in myset
```

```
Out[58]: True
```

```
In [59]: 'ten' in myset
```

```
Out[59]: False
```

```
In [64]: #add and remove items  
myset.add('NINE')  
myset
```

```
Out[64]: {'NINE', 'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [66]: myset.update(['TEN','ELEVEN','TWELVE'])  
myset
```

```
Out[66]: {'ELEVEN',
          'NINE',
          'TEN',
          'TWELVE',
          'eight',
          'five',
          'four',
          'one',
          'seven',
          'six',
          'three',
          'two'}
```

```
In [67]: myset.remove('NINE')
myset
```

```
Out[67]: {'ELEVEN',
          'TEN',
          'TWELVE',
          'eight',
          'five',
          'four',
          'one',
          'seven',
          'six',
          'three',
          'two'}
```

```
In [69]: myset.discard('TEN')
myset
```

```
Out[69]: {'ELEVEN',
          'TWELVE',
          'eight',
          'five',
          'four',
          'one',
          'seven',
          'six',
          'three',
          'two'}
```

```
In [70]: myset
```

```
Out[70]: {'ELEVEN',
          'TWELVE',
          'eight',
          'five',
          'four',
          'one',
          'seven',
          'six',
          'three',
          'two'}
```

```
In [71]: myset.clear()# Deleteall items ina setmyse
myset
```

```
Out[71]: set()
```

```
In [74]: del myset# Delete the set objectmyset
myset
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[74], line 1
----> 1 del myset# Delete the set objectmyset
      2 myset

NameError: name 'myset' is not defined
```

```
In [75]: #copy set
myset={'one','two','three','four','five','six','seven','eight'}
myset
```

```
Out[75]: {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [76]: myset1=myset# Create a new reference "myset1"myset1
```

```
In [77]: id(myset),id(myset1)#The addressof both myset& myset1 willbe thesame as previous
```

```
Out[77]: (1809749979904, 1809749979904)
```

```
In [78]: my_set=myset.copy()#Create a copyof the listmy_set
```

```
In [79]: id(myset)
```

```
Out[79]: 1809749979904
```

```
In [80]: myset
```

```
Out[80]: {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [81]: list(enumerate(myset))
```

```
Out[81]: [(0, 'six'),
          (1, 'three'),
          (2, 'seven'),
          (3, 'four'),
          (4, 'eight'),
          (5, 'five'),
          (6, 'two'),
          (7, 'one')]
```

```
In [82]: #union
A={1,2,3,4,5}
B={4,5,6,7,8}
C={8,9,10}
```

The history saving thread hit an unexpected error (OperationalError('database or disk is full')).History will not be written to the database.

```
In [83]: A|B
```

```
Out[83]: {1, 2, 3, 4, 5, 6, 7, 8}
```

```
In [84]: A.union(B,C)
```

Out[84]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

```
In [86]: A.update(B,C)  
A
```

Out[86]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}